DAY 25 | HANDS-ON ACTIVITIES

SLMs and Token Optimisation: Problem Statements

Seven problems. Every notebook cell runs offline; add an OPENAI_API_KEY to see real model behaviour. Offline results show how the code works, not real model quality. All prices are placeholders.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: SLM or LLM? NEW (v4)

Level: Warm-up | **Concepts:** model choice, latency, privacy

For ten SmartDesk jobs (for example: tag ticket language, classify category, draft a refund reply, summarise a 40-minute call, detect abusive text, plan a fibre rollout), choose SLM or LLM and give a one-line reason using the four questions: how hard, how often, how private, how fast.

Sample run / expected result

```
tag language      -> SLM: narrow, very high volume
plan fibre rollout -> LLM: open-ended, multi-step, low volume
```

Hints

- There is no single right answer. The reason matters more than the choice.

Problem 2: Memory Footprint NEW (v4)

Level: Easy | **Concepts:** quantisation

Write weights_gb(params_billion, bits) and print a table of weight memory for 0.5B, 1.5B, 3B and 7B models at 16, 8 and 4 bits.

Sample run / expected result

```
params  fp16   int8   int4
0.5B    1.0    0.5    0.25
3.0B    6.0    3.0    1.5
7.0B   14.0    7.0    3.5
```

Hints

- bytes = params x bits / 8. Weights only; runtime memory is higher.

Problem 3: SLM vs LLM on 20 Tickets NEW (v4)

Level: Medium | **Concepts:** accuracy, latency, cost

Run the notebook comparison on the 20 labelled SmartDesk tickets. Report accuracy, average latency and estimated cost per 1,000 tickets for both models, then list the tickets each model got wrong.

Sample run / expected result

```
model   accuracy  avg_ms  cost_per_1k
slm       0.85      ...     ...
llm       0.90      ...     ...
```

Hints

- Use the same prompt and label set for both.
- Offline numbers come from stand-ins; say so.

Problem 4: Token Bill per Feature NEW (v4)

Level: Medium | **Concepts:** input, output, cached, reasoning tokens

Using the placeholder prices, compute the cost per 1,000 requests for three SmartDesk features: ticket summary (600 in, 60 out), FAQ answer (1,500 in, 300 out) and escalation plan with reasoning (2,000 in, 400 out, 3,000 reasoning).

Sample run / expected result

```
summary     : 0.84 per 1,000
faq answer  : 2.70 per 1,000
plan        : 15.60 per 1,000
```

Hints

- Reasoning tokens are billed as output.

Problem 5: Caching and Trimming NEW (v4)

Level: Medium | **Concepts:** prompt caching, context trimming, output limits

Start from the FAQ answer feature. Apply 50% cached input, then trim context from 500 to 250 tokens and cap output at 150. Print the cost per 1,000 after each step and the total saving.

Sample run / expected result

```
baseline : 2.70
+ cache  : 2.02
+ trim   : 1.18
saving   : 56%
```

Hints

- Change one thing at a time so you know which lever saved what.

Problem 6: Route SLM First NEW (v4)

Level: Challenging | **Concepts:** model routing, escalation

Write route(ticket) that uses the SLM classifier and escalates to the LLM when confidence is below a threshold. Try thresholds 0.5, 0.7 and 0.9 and report escalation rate, accuracy and cost per 1,000.

Sample run / expected result

```
threshold  escalated  accuracy  cost_per_1k
0.5          10%        ...       ...
0.9          45%        ...       ...
```

Hints

- A higher threshold escalates more: better accuracy, higher cost.

Problem 7: FinOps Guard NEW (v4)

Level: Stretch | **Concepts:** budgets, quotas, daily usage

Build a small usage tracker: per-user daily token quota, a project monthly budget, and alerts at 50%, 80% and 100%. Feed it a week of simulated Codex and app usage and print the alerts raised.

Sample run / expected result

```
day 3: user ana reached 80% of daily quota
day 5: project at 50% of monthly budget
```

Hints

- Keep Codex and app usage separate, as in the access plan.